

WITCHCLIFFE, Australia

WMO: 956410

Lat: 34.0281S

Lon: 115.1042E

Elev: 81

StdP: 100.36

Time Zone: 8.00 (E08)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.9	5.1	2.3	4.5	8.2	3.1	4.8	9.0	10.6	15.8	10.0	15.6	0.2	150	0.501

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	11.1	31.3	19.5	29.2	19.0	27.1	18.4	20.8	27.6	19.9	26.0	19.2	24.7	5.0	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.2	14.1	21.4	18.2	13.3	20.6	17.4	12.6	20.4	60.4	27.9	57.4	26.1	55.1	24.7	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.8	9.1	8.1	1.5	36.2	1.3	1.5	0.6	37.3	-0.2	38.1	-0.9	39.0	-1.8	40.0
			1.6	22.3	1.0	0.9	0.9	23.0	0.3	23.5	-0.2	24.0	-1.0	24.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.6	19.6	20.0	18.8	16.3	14.2	12.8	12.0	12.2	12.6	14.0	16.3	18.1
	DBStd	3.72	2.63	2.84	2.82	2.41	1.98	1.99	2.06	1.86	1.92	2.18	2.59	2.72
	HDD10.0	11	0	0	0	0	1	1	5	3	2	0	0	0
	HDD18.3	1222	14	12	28	69	127	166	197	191	173	137	73	36
	CDD10.0	2039	298	281	272	191	132	85	67	71	79	123	189	252
	CDD18.3	207	53	59	42	9	1	0	0	0	0	2	11	31
	CDH23.3	1656	407	462	325	66	6	0	0	0	1	19	117	254
	CDH26.7	529	133	156	113	11	0	0	0	0	0	2	33	82
(14) Wind	WSAvg	3.8	5.0	4.7	3.8	3.1	2.9	3.1	3.3	3.4	3.8	3.9	4.2	4.3
(15) Precipitation	PrecAvg	1040	14	12	32	60	147	195	200	155	110	61	35	20
	PrecMax	1307	63	57	113	144	248	293	377	220	208	126	104	59
	PrecMin	778	0	1	8	6	56	103	84	44	39	22	0	4
	PrecStd	142	17	13	23	37	48	59	65	35	42	27	24	14
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.7	33.5	34.0	28.2	24.0	19.9	18.9	19.7	21.8	25.8	30.5	33.1
		MCWB	20.0	20.5	20.0	17.6	15.3	14.9	15.4	14.9	14.4	17.5	17.7	19.0
	2%	DB	30.2	30.8	29.7	25.2	21.5	18.7	17.5	18.2	19.2	22.6	26.7	29.1
		MCWB	19.8	19.9	19.0	17.3	15.7	14.6	14.3	14.4	14.8	16.4	17.3	18.6
	5%	DB	27.5	28.5	26.6	23.0	20.0	17.8	16.7	17.2	17.8	20.2	23.7	26.1
		MCWB	18.7	19.0	18.4	17.1	15.6	14.1	13.6	13.8	14.2	15.2	16.6	18.1
	10%	DB	25.2	26.1	24.3	21.4	18.8	16.9	15.7	16.2	16.7	18.4	21.6	23.8
		MCWB	18.0	18.5	17.7	16.4	14.9	13.7	12.9	13.1	13.2	14.4	15.8	17.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.0	22.4	21.2	19.7	18.6	16.7	16.8	16.8	17.3	18.4	19.4	20.8
		MCDB	28.9	28.3	28.1	23.0	20.5	17.9	17.5	17.9	19.1	23.7	26.6	28.7
	2%	WB	20.6	20.8	20.1	18.7	17.3	15.9	15.3	15.6	15.8	17.0	18.2	19.4
		MCDB	27.2	27.6	25.2	22.2	19.5	17.5	16.7	17.2	18.0	20.9	23.4	26.5
	5%	WB	19.6	19.8	19.3	17.9	16.3	15.0	14.2	14.5	14.5	15.9	17.4	18.4
		MCDB	25.6	26.4	24.0	21.3	18.6	16.9	15.9	16.5	17.1	18.9	21.9	24.2
	10%	WB	18.6	19.0	18.5	17.1	15.5	14.2	13.3	13.5	13.6	15.0	16.5	17.5
		MCDB	24.2	24.6	23.1	20.4	18.0	16.2	15.2	15.6	16.1	17.6	20.8	22.5
(35) Mean Daily Temperature Range	5% DB	MDBR	10.9	11.1	10.7	9.9	9.2	7.6	7.0	7.4	7.7	8.6	9.9	10.4
		MCDBR	15.3	15.2	15.2	13.1	11.7	9.0	8.0	9.4	10.4	12.6	14.3	14.6
		MCWBR	6.2	6.0	6.2	6.3	7.0	6.1	5.5	5.9	6.5	6.8	6.5	6.0
	5% WB	MCDBR	12.8	12.6	11.7	10.2	8.9	7.2	6.7	7.6	8.7	10.2	11.9	12.4
		MCWBR	5.5	5.3	5.4	5.9	6.4	5.8	5.6	5.8	6.2	6.2	6.1	5.5
(40) Clear-Sky Solar Irradiance	taub		0.332	0.332	0.322	0.317	0.312	0.306	0.303	0.314	0.327	0.324	0.328	0.327
	taud		2.599	2.609	2.638	2.646	2.628	2.626	2.634	2.590	2.534	2.557	2.572	2.596
	Ebn at Noon		1005	985	960	911	860	839	862	899	938	981	1003	1014
	Edh at Noon		103	99	89	79	71	67	70	82	97	103	105	104
(44) All-Sky Solar Radiation	RadAvg		7.68	6.83	5.43	3.88	2.82	2.25	2.43	3.28	4.43	5.83	7.02	7.67
	RadStd		0.40	0.29	0.32	0.22	0.18	0.15	0.11	0.18	0.23	0.33	0.38	0.38

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page